Drawback of traditional model :-

* Customer invest the money on day 1, return of investment takes a lot of time.
* People will not be responsible.
* Customer give the requirement on day 1, & will not be in contact till the end.
* the products will not the market demands.

Agile model

* It is an incremental & iterative process to develop the software
* to overcome drawback of traditional model we come to agile model
* Here we develop a large software in short cycle called "sprint".

Not including in Agile methodologies

1. Scrum process
2. Extreme progress
3. Lean & kanban
4. crystal
5. Dynamic s/w development method [DSDM]
6. Adaptive —"— (ASDM)
7. FDD|TDD

Scrum process :-

* It is the process to develop s/w in Agile model

Scrum team :-

* It is a team which consist of 7 to 12 Engineers working towards completing the committed features
* It includes two team "core team & shared team"
* The member of core team is {developers, testers, scrum master}
* The member of shared team is business analyst, UI-UX, product owner, Developers Engineer, N/w & db admin.
* the Scrum master facilitates Everyone to complete there tasks.

* **product Backlog :-**
* It is a list of prioritized stories which should be completed in the complete/entire project
* Generally be a product owner, customer architect and scrum master will be involved in building it.
* product backlog need not be in detail.

sprint Backlog:-
It is a list of prioritized stories and associated tasks which has to completed in one sprint.

1. sprint planning meeting :-
* Here the entire scrum team will sit together and pull the stories from product backlog.
* the scrum master will assign the stories to the developers and test engineers.
* now the engineers will derive the task which has to completed in order to build each story.
* They will estimate the time taken to complete each task it is called as "story point".
* This meeting should be completed within 8hrs (2hrs/week)

2. Daily standup meeting | Rollcall meeting | Daily scrum meeting
* This meeting is conducted by "scrum master"
* Here all the engineers will talk about
  a) What they have done yesterday
  b) What were the "impediments | hurdle" faced yesterday.
  c) What are they going to do today
  d) What are impediments | hurdle they are expecting today

* the Scrum master tries to solve the problem then and there, if it take a lot of time he notes it in itself, if it take a lot of time he notes it in "impediment backlog" & he will solve it later.

* This meeting takes only 10 to 15 minutes

* It should be conducted in the begining of the day

* Everyone in this meeting should stand and talk so that they speak to the point.

Sprint Review meeting :

* Here the Engineers will give demo to the product owner.
* the product will say what is done & what is ~~done~~ not done.
* Here they also plan for upcoming sprint.

Retrospective meeting :

* Here the Entire Scrum team meets
* they will discuss what are the achivments & mistakes they have done in the current sprint.
* They will document it & this document is called "Retrospect document"
* they will refer this document in upcoming sprint so that the achivments should be followed & mistakes should be avoided.

Burndown chat :

* It is a graphical representation of workleft v/s time

Story board / white board :

It is a list of pending task, ~~and~~ task inprogress and completed task.

Acceptance Criteria

* It is a criteria which should be met in order to move the software to the production (or) to customer.
* It is usually set by product owner / customer.

Hot fix / Incident management

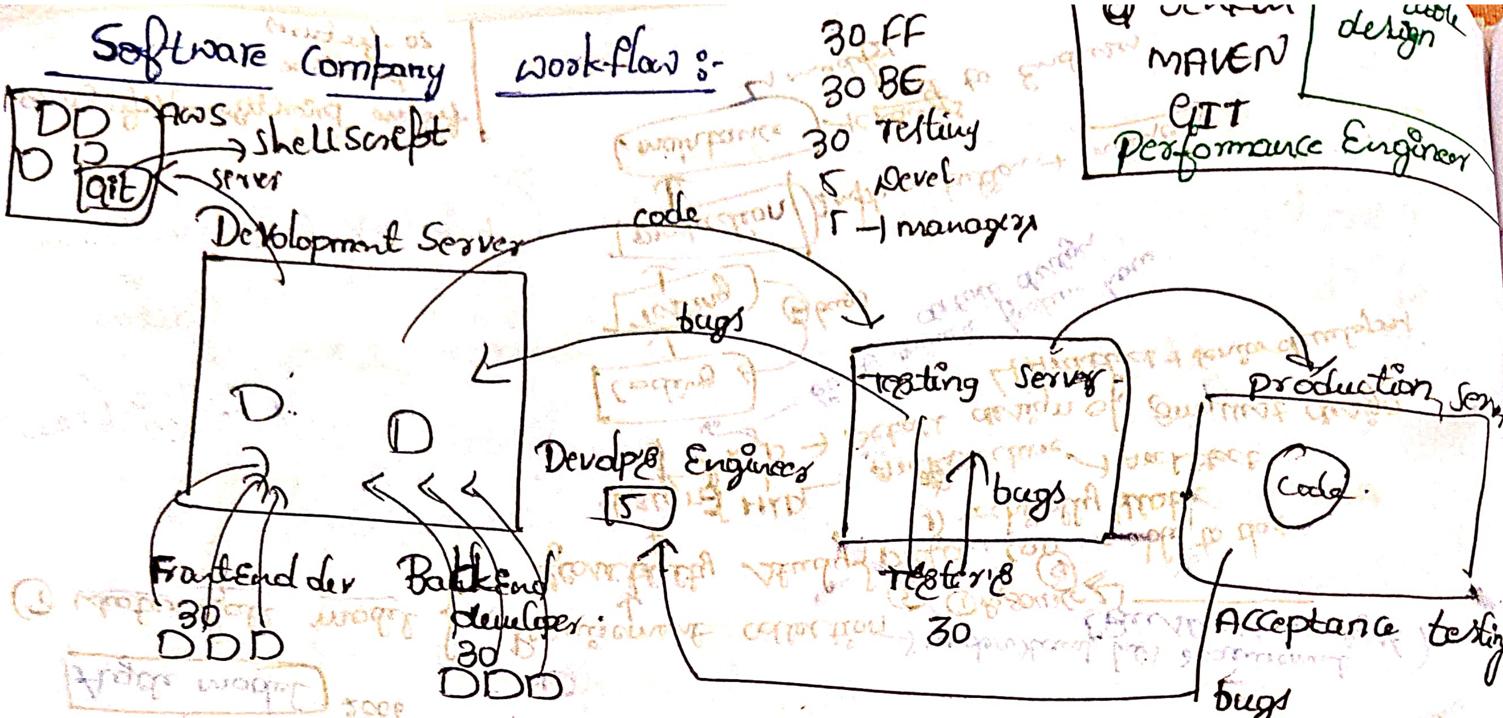
when the software is moved to production, if customer finds and blocker / ~~or~~ critical defects it will be communicated to the software company.

* the developers will immediately fix the defect the test Engineers will retest and patch will be created.
* this patch will be installed in the production server
* It might take <u>3hrs to 3days</u>.

<u>Chicken :-</u>

* It is a person who will observe the task but will not be involved in any task. { like stakeholder, investors }

Software Company workflow :-



Devops Engineer works
1) Maintaince the clean Code
2) Send the Code to testing server
4) Take a bugs from acceptance testing and manage.
5) Send to production server

unit testing :- chks the code line by line

Types of testing :-

1. functionality Testing – Testing Each & Every Component
2. Integration Testing → A to B { Request msg accept msg (facebook)
3. System Testing → End to End testing { All kinds }
4. Acceptance testing → another level testing done by client
5. Unit testing – testing every line of code by developers (white box testing) { Junit, test Engineer, Pytest }

- Backlog :-
  1. Product backlog
  2. sprint backlog
  3. Impediment Backlogs

Software development life Cycle [SDLC]

LDA Server
↳ username | password
stored in one db

It is an procedure to develope a Software

1) Waterfall model (1970)
2) Spiral
3) V-model

Agile model 2006

① Water fall model :-

Requirement Collection → {understand requirement [Business analyst]}

↓

feasibility Study → technically viable (able to do it
→ profit
→ resources

↓

design → {HLD, LLD}

HLD → full architecture (Ex:- Banking → Architect
saving, current, ...
↳ p

LLD → detail design of
Smallest design {Architect + Senior develop

↓

Coding → ~~full~~ make structure to (HLD & LLD)

↓

Testing → testing for bugs

↓

Production } implementation (End user)

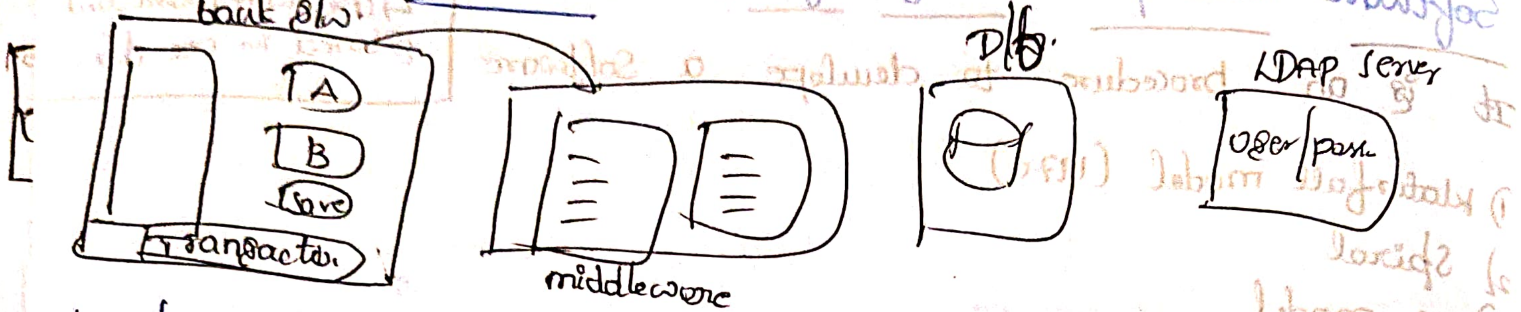
↓

maintance {change need as per End user}

Drawback :-

1) inflexible (Hard to make changes
2) High risk → Error can costly to fix
3) long development time (2 year - 3 years)
4) Limited customer feedback {customer may not see that product until it's finished}

HLD (High level design)

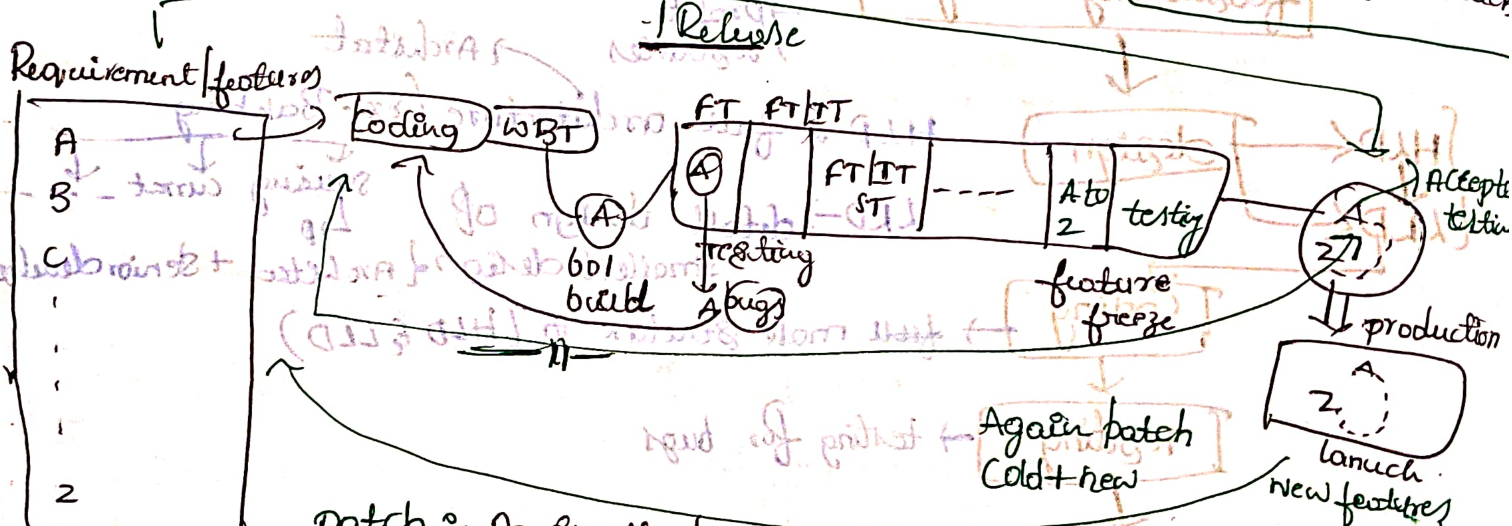


Low level design

Transfer.java
→ check block user.java
→ check balance.java
→ deduct Add.java
→ sent msg.java

feature freez :- stop adding feature & fix the bug

1 Release Starting from requirement to production is 1 realease / launch

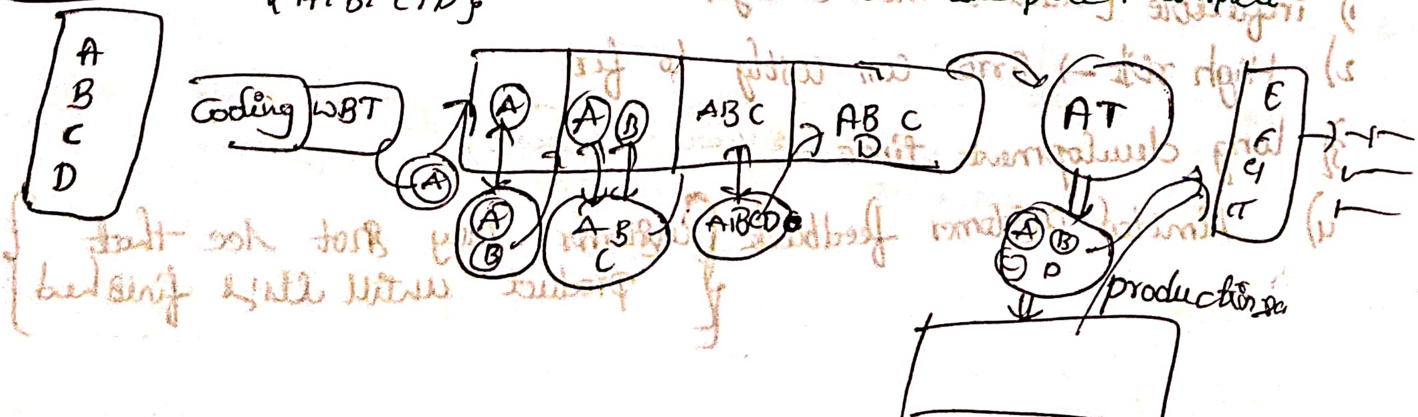


patch : A small s/w it contain new program + updated program + old program.
↙ develop's Engineer

1. During the 2nd time adding the features take a copy of first release & do it.
2. patching / updates.

Agile model {development of product in less time}

1 sprint :- It takes 1-2 months from requirements to productions (launch)
{A, B, C, D}
↳ make piece & complete



Agile model:- It is an iterative & flexible approach to develope a project & slow development & shorter cycle

Integration testing:- Testing dataflow b/w two modules.

(1) Requirements Collection:-

prioritize ─ technically
└→ Business

$\boxed{\text{Product backlog}}$ → leftout features to build
↳ ~~to~~ prioritize list ~~out~~ stories that must build in products given from customer.

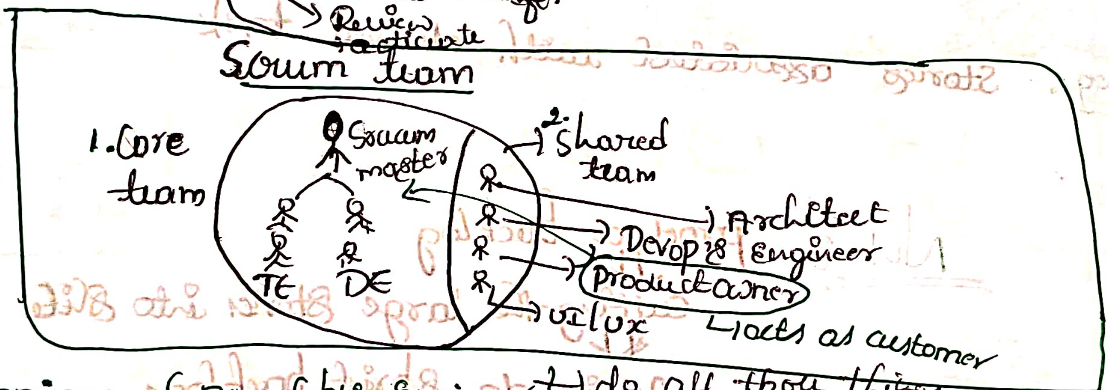
Each piece of requirements called Story

(2) Loan
- home → (1) {Apply, Approval} → Story
- personal (2)
- vehicle (3)
- mortage

(3) Insurance
- health (3)
- life → (1)
- vehicle (2)

(1) Core banking
- create account → (1)
- Approve account
- Amount transfer
- Review / activate

} product backlogs
↳ priortize list of stories which must be developed Entire produ developer.



Sprint
(2) planning:- (Develop's engineer do all these things)
* choose stories from product backlogs
* Assign the stories to developers.
* priortize the stories {which story is develope first
* Derive the task {LLD, db design, UI/UX, coding, unit testing} — 8hr, 44hrs, 48hrs, 120hrs, 48hrs
* prioritize the task {UI/UX, DB design, LLD, coding & testing}
* Estimate / derive the story points (times)
* ~~Derive~~ the ~~stories~~ points

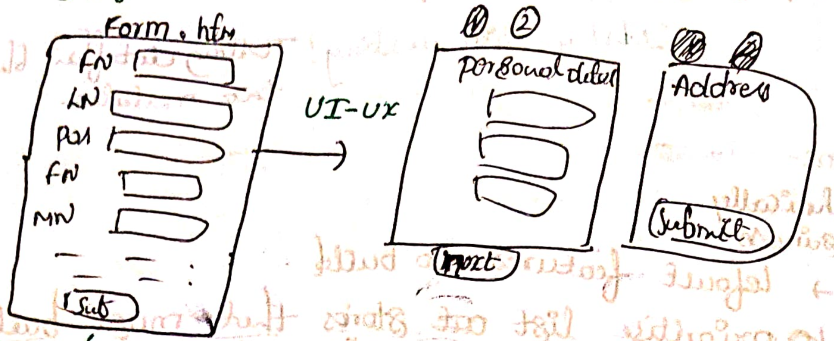
} Sprint planning

* Note:- Sprint is 4weeks ⇒ do a planning in 8hrs

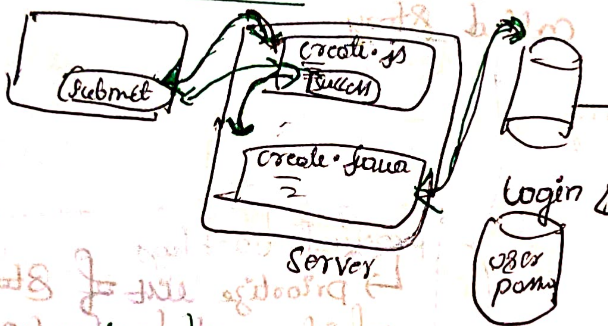
Step 1:- product owner priortize the stories
2:- Scrum master assign to developers, give a helps to developers
3:- product owner check for acceptance Criteria

UI :- looking interactive to user

UX :- make sure user Experience good Ex :- long form filling use next page



Database design :-



heavy data so create separate table for login so it will be done very fast.

→ UI-UX first
→ Database ②
→ LLD
→ coding

flow must follow

Developers work :-
fix the bugs of old feature + add new features

Sprint Backlog :- Stories associated with derived task

Corebanking	develop Assign	Prioritize	Derive task	Priority task	estimate
Create A/C		①	LLD	③	6
Review A/C		②	design db	②	4
Approve A/C		③	UI/ux	①	8
Activate A/C		④	coding	④	20
			unit testing	⑤	4

Sprint backlog

Note :- product backlog
↓↓
cutting the large Stories into Slite
↓↓
Again do sprint backlogs

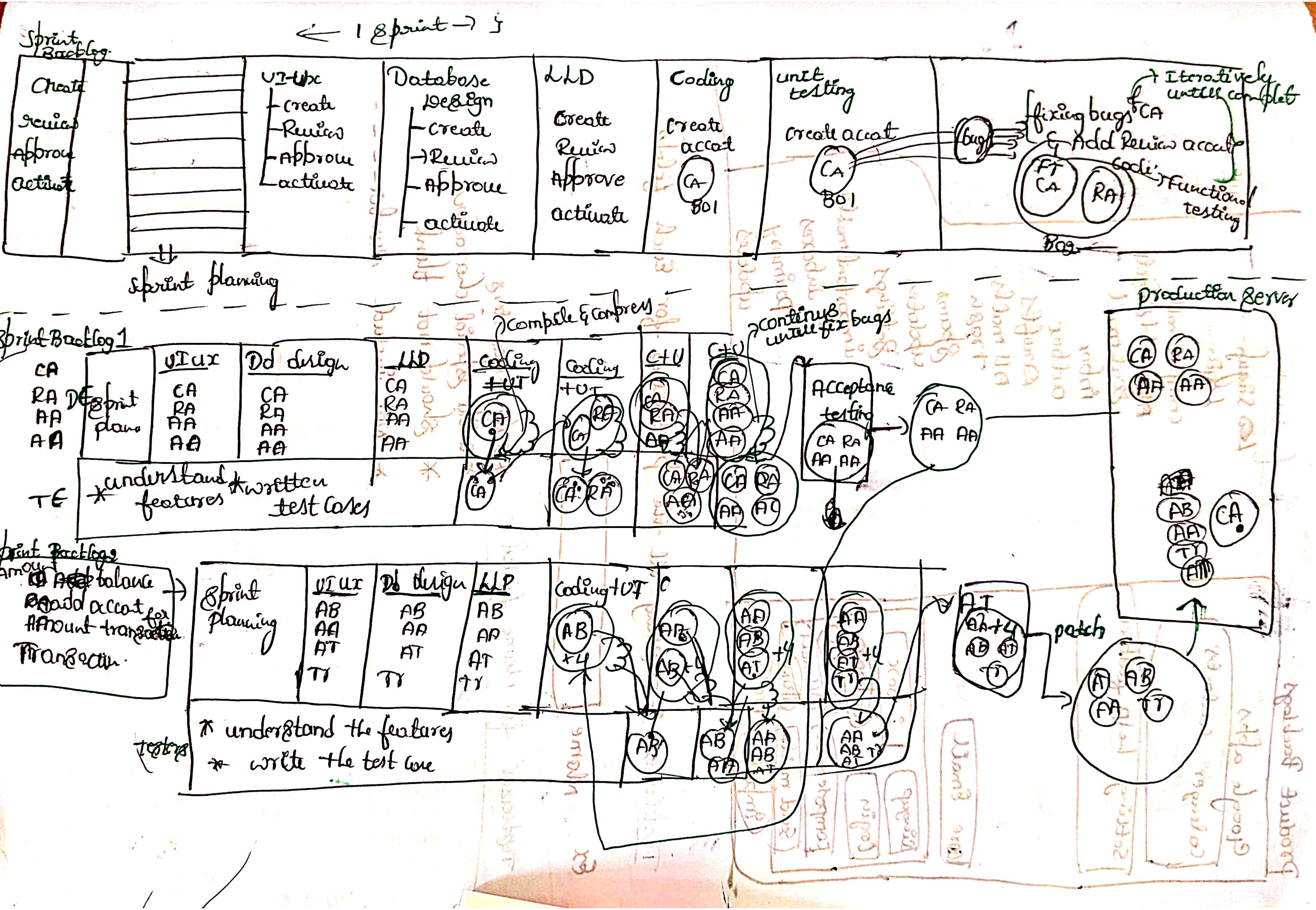
Product Backlogs

Loan

Insurence

Core Banking ①

Create A/C
Review
Aprove
Activate



← 1 Sprint →
Sprint Backlog
Create
review
Approve
Activate
UI-Ux
create
Review
Approve
activate
Database Design
create
Review
Approve
activate
LLD
Create
Review
Approve
activate
Coding
Create accout
CA
B01
unit testing
Create accout
CA
B01
bugs
fixing bugs CA
& Add Review accout
Iteratively untill complet
FT
CA
RA
Code → Functional testing
Bag
Sprint planing
Sprint Backlog 1
CA
RA
AA
AA
Sprint plans
UI UX
CA
RA
AA
AA
Dd design
CA
RA
AA
AA
LLD
CA
RA
AA
AA
Coding +UT
Coding +UT
C+U
C+U
compile & compress
continus untill fix bugs
Acceptance testing
CA RA
AA AA
CA RA
AA AA
TE
* understand features
* written test cases
Production Server
CA RA
AA AA
AB
AA
TY
AM
CA
Sprint Backlog 2
Amount balance
add accout for Amount transaction
Transaction
Sprint planning
UI UX
AB
AA
AT
TY
Dd design
AB
AA
AT
TY
LLD
AB
AA
AT
TY
Coding+UT
C
AB
+4
AT
AA+4
AB
AT
patch
AT
AA
AB
TY
testers
* understand the features
* write the test case

Product Backlogs

Google apps

Calenger contacts

Setting help feedback

Core Email 1

Singup
inbox
Login
Outbox
Drafts
compose mail
All mail
Send mail
Spam
Important
Trash

features

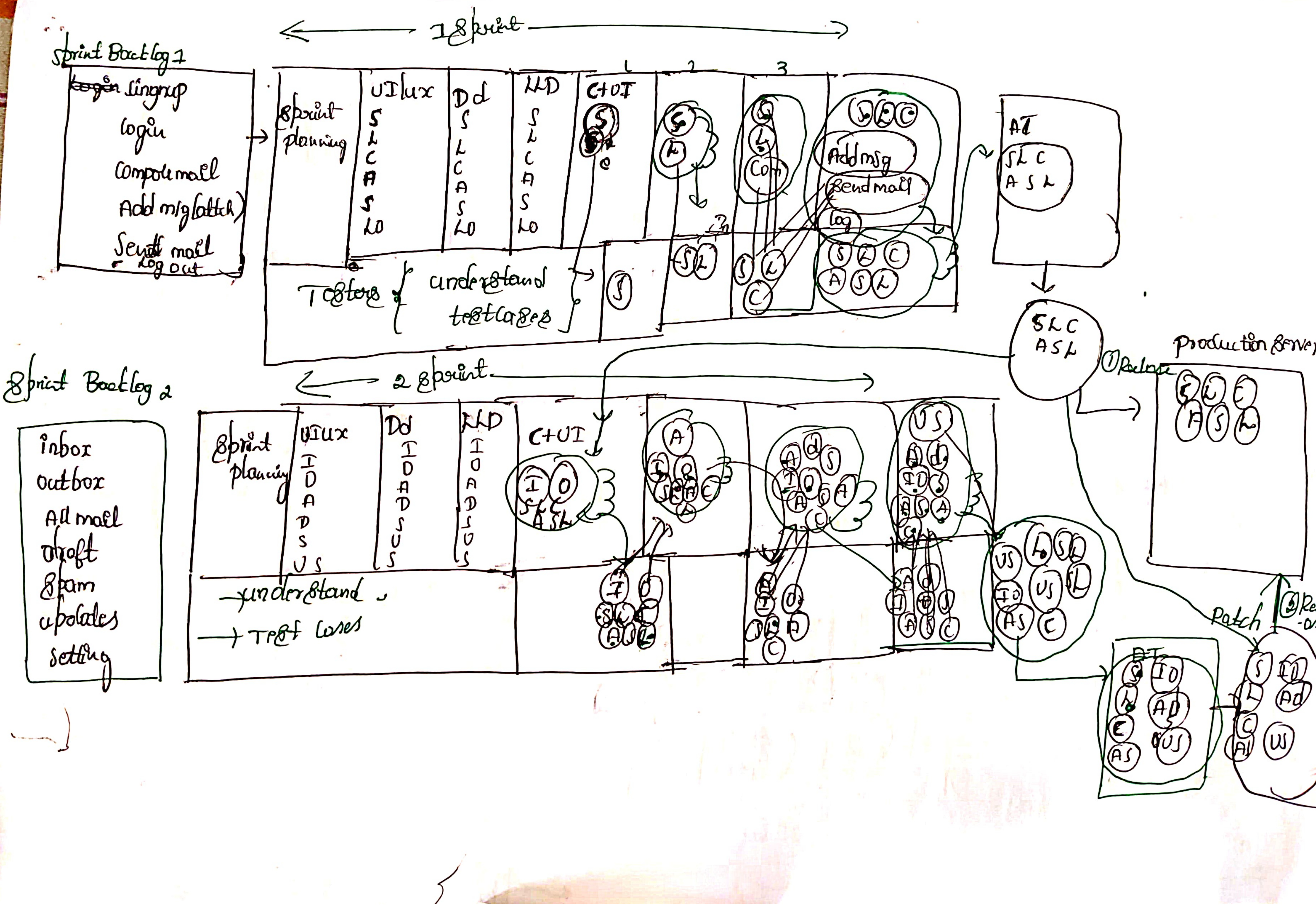
~~Log~~ Singup → 1
Login → 2
compose mail → 3
~~attach~~ add Subject → 4
Send mail → 5
} 1 Sprint
inbox → 6
outbox → 7
Drafts → 8
All mails → 9
} → 2
trash
Spams → 10
updates
Settings
important mails
All inboxes
Primary
updates

Test Cases :- Testing all the possibilities for Each fields

Ex Name [ ]

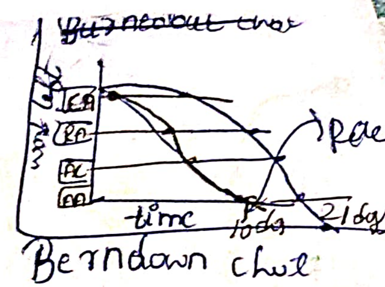
Testcases for Name field :-

* only characters
* avoid special characters
* should not blank
* not include number - - - -


1 Sprint
Sprint Backlog 1
~~login~~ singup
login
compose mail
Add msg(attch)
Send mail
log out
Sprint planning
UI/ux
S L C A S LO
Dd
S L C A S LO
LLD
S L C A S LO
C+UI
1
2
3
Add msg
Send mail
log
Com
Testers
understand
test cases
AT
SLC ASL
Release
Production Server
2 Sprint
Sprint Backlog 2
inbox
outbox
All mail
draft
Spam
updates
setting
UIUX
I D A D S U S
Dd
I D A D S U S
LLD
I D A D S U S
C+UI
understand
Test cases
Patch
Re
DT

1) ~~Sprint planning~~
2) Standup meeting
~~3) Impediment Backlog~~
3) Sprint Review meeting
4) Retrospective meeting

} 4 Ceremony



10-minutes

developers

- Standup meeting → Yesterday ✓✓✓ X X (I did / didn't)
- daily standup call → Today ✓✓ → It will do this, I should do but i can't some reasons
- Scrum call
- daily scrum
- Daily Rollcall

} Scrum master Note down obstacle X

3) Impediment Backlog:- → List of the problem not solved todays to do it

4) Sprint Review meeting :- → check acceptance criteria}

→ Feature Demo (present to product owner, Scrum master, customers ---)

5) Retrospecting meeting → After 1 Release before starts another Sprint { good things, mistakes }